

Volumes of Revolution

ANSWER KEY



The disk and washer methods

- Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x -axis.
$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = 8\pi.$$
- The region between $y = x^2$ and $y = 4$ is revolved about the x -axis. Using the washer method, set up and evaluate the volume integral.
Bounds $x = -2$ to 2 :
$$V = \pi \int_{-2}^2 [4^2 - (x^2)^2] dx = \pi \int_{-2}^2 (16 - x^4) dx = \pi \left[16x - \frac{x^5}{5} \right]_{-2}^2 = \frac{384\pi}{5}.$$

The shell method

- The region under $y = x^2$ on $[0, 2]$ is revolved about the y -axis. Use the shell method to find the volume.
$$V = 2\pi \int_0^2 x \cdot x^2 dx = 2\pi \int_0^2 x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^2 = 8\pi.$$
- The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y -axis. Find the volume using the shell method.
$$V = 2\pi \int_0^3 x \cdot x dx = 2\pi \left[\frac{x^3}{3} \right]_0^3 = 18\pi.$$

Comparing methods

- The region under $y = x^2$ on $[0, 2]$ is revolved about the x -axis (not the y -axis). Find the volume using the disk method.
$$V = \pi \int_0^2 (x^2)^2 dx = \pi \int_0^2 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^2 = \frac{32\pi}{5}.$$
- The region bounded by $y = \sqrt{x}$ and $y = x$ (between their two intersection points) is revolved about the x -axis. Find the volume using the washer method.
Intersections: $\sqrt{x} = x$ at $x = 0, 1$. On $[0, 1]$, $\sqrt{x} \geq x$, so the outer radius is $\sqrt{x}$, inner is x :
$$V = \pi \int_0^1 [(\sqrt{x})^2 - x^2] dx = \pi \int_0^1 (x - x^2) dx = \pi \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{\pi}{6}.$$

Volumes about other axes

- The region under $y = x^2$ on $[0, 2]$ is revolved about the line $y = -1$. Find the volume, using outer radius $x^2 + 1$.
$$V = \pi \int_0^2 [(x^2 + 1)^2 - 1^2] dx = \pi \int_0^2 (x^4 + 2x^2) dx = \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} \right]_0^2 = \pi \left(\frac{32}{5} + \frac{16}{3} \right) = \frac{176\pi}{15}.$$
- The region under $y = x^2$ on $[0, 2]$ is revolved about the line $x = 3$ (using the shell method with radius $3 - x$).
$$V = 2\pi \int_0^2 (3 - x)x^2 dx = 2\pi \int_0^2 (3x^2 - x^3) dx = 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^2 = 2\pi(8 - 4) = 8\pi.$$

Volumes with trigonometric and exponential curves

- 9 The region under $y = \sin x$ on $[0, \pi]$ is revolved about the x -axis. Find the volume, using $\sin^2 x = \frac{1 - \cos 2x}{2}$.
- $$V = \pi \int_0^\pi \sin^2 x \, dx = \pi \int_0^\pi \frac{1 - \cos 2x}{2} \, dx = \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^\pi = \frac{\pi}{2} (\pi - 0) = \frac{\pi^2}{2}.$$
- 10 The region under $y = e^{-x}$ on $[0, 1]$ is revolved about the x -axis. Find the volume, to three decimal places.
- $$V = \pi \int_0^1 e^{-2x} \, dx = \pi \left[-\frac{1}{2} e^{-2x} \right]_0^1 = \frac{\pi}{2} (1 - e^{-2}) \approx \frac{\pi}{2} (0.8647) \approx 1.358.$$

Setting up without evaluating

- 11 Set up (do not evaluate) the volume integral for the region under $y = \ln x$ on $[1, e]$, revolved about the x -axis.
- $$V = \pi \int_1^e (\ln x)^2 \, dx.$$
- 12 Set up (do not evaluate) the volume integral for the region between $y = x^2$ and $y = \sqrt{x}$ (on $[0, 1]$) revolved about the y -axis, using the shell method.
- $$V = 2\pi \int_0^1 x(\sqrt{x} - x^2) \, dx \text{ (outer curve } \sqrt{x} \text{ minus inner curve } x^2, \text{ since } \sqrt{x} \geq x^2 \text{ on } [0, 1]).$$

Visualizing the generating region

- 13 The region under $y = \sqrt{x}$ on $[0, 4]$ (from an earlier disk-method problem) is shown. Sketching this region mentally revolved about the x -axis produces a solid — describe its shape, and confirm the volume 8π found earlier is reasonable by comparing to a cylinder of radius 2 and length 4.
- The solid is a paraboloid-like horn, narrow at $x = 0$ and widening to radius 2 at $x = 4$. A full cylinder of radius 2, length 4 would have volume $\pi(2)^2(4) = 16\pi$; since the paraboloid is narrower than the cylinder everywhere except at $x = 4$, a volume of 8π (exactly half the cylinder) is reasonable.

Volume of a solid with a hole (washer method practice)

- 14 The region between $y = x$ and $y = x^3$ on $[0, 1]$ is revolved about the x -axis. Find the volume using the washer method (outer radius x , inner radius x^3).
- $$V = \pi \int_0^1 [x^2 - x^6] \, dx = \pi \left[\frac{x^3}{3} - \frac{x^7}{7} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{7} \right) = \frac{4\pi}{21}.$$

An applied volume problem

- 15** A vase is modeled by revolving the region under $y = \sqrt{x} + 1$ on $[0, 4]$ about the x -axis, where x is the height (cm) and y is the radius. Find the volume of the vase.

$$V = \pi \int_0^4 (\sqrt{x} + 1)^2 dx = \pi \int_0^4 (x + 2\sqrt{x} + 1) dx = \pi \left[\frac{x^2}{2} + \frac{4}{3}x^{3/2} + x \right]_0^4 = \pi \left(8 + \frac{32}{3} + 4 \right) = \pi \left(\frac{68}{3} \right) = \frac{68\pi}{3} \text{ cm}^3.$$

Volume with the shell method, revisited

- 16** The region bounded by $y = 4 - x^2$ and $y = 0$ (for $x \geq 0$) is revolved about the y -axis. Use the shell method to find the volume.

$$V = 2\pi \int_0^2 x(4 - x^2) dx = 2\pi \int_0^2 (4x - x^3) dx = 2\pi \left[2x^2 - \frac{x^4}{4} \right]_0^2 = 2\pi(8 - 4) = 8\pi.$$